MADSON ALLAN DE LUNA ARAGÃO

Bioinformatics Engineer, Software Development & Data Analyst

Belo Horizonte, MG, Brazil
madsondeluna@gmail.com
Portfolio | GitHub | LinkedIn | Google Scholar | ORCID

PROFESSIONAL SUMMARY

Bioinformatics Engineer and scientific software developer with experience in research and the development of solutions integrating software engineering, machine learning, omics data analysis and integration, structural biology and chemoinformatics. Combines sequence analysis, genomics, transcriptomics, molecular modeling and molecular dynamics simulations to investigate biological mechanisms, characterize molecules and guide candidate discovery and optimization. Develops reproducible pipelines, web applications and command-line tools using Python, R and Linux and high-performance computing environments, with Slurm, Nextflow and Snakemake.

Designs and computationally optimizes synthetic or bioinspired peptides and proteins using deep learning, generative models and protein language models to develop candidates tailored to specific functions, with potential diagnostic, therapeutic and vaccine applications. Integrates sequence information, physicochemical properties and structural evidence into candidate generation, assessment and prioritization. Current research focuses on developing antimicrobial peptides and investigating their interactions with membranes, connecting molecular engineering and artificial intelligence to the discovery of new strategies against antimicrobial resistance.

Research and laboratory experience spans the Oswaldo Cruz Foundation (Fiocruz), the Federal University of Pernambuco (UFPE) and its university hospital (HC-UFPE), alongside current research at the Federal University of Minas Gerais (UFMG). Holds a master's degree in Genetics and Molecular Biology from UFPE and is pursuing a PhD in Bioinformatics at UFMG. Currently serves as Bioinformatics Engineer and Head of Data Science at Biotech Business and as a Nextflow Ambassador, contributing to the development and improvement of scientific tools and workflows.

TECHNICAL SKILLS

- **Genomics and multi-omics integration:** genome and proteome mining, identification and annotation of uncharacterized genes, sequence analysis, functional annotation, and integration of genomic, transcriptomic and structural evidence for candidate characterization and prioritization.
- **Transcriptomics and RNA-seq:** experimental and computational experience, gene expression analysis, interpretation of RNA-seq data, and integration of results with functional and structural information to inform hypotheses and scientific decisions.
- **Molecular modeling and simulation:** protein, peptide and membrane modeling; molecular docking, classical molecular dynamics, trajectory analysis, stability, flexibility and molecular interactions. GROMACS, HADDOCK, PyMOL, MDAnalysis and Rosetta.
- **Peptide and protein design:** de novo, target-guided and function-specific design, sequence optimization and structural assessment of candidates for potential diagnostic, therapeutic and vaccine applications. Integration of physics- and energy-based methods with deep learning. Rosetta, RFdiffusion, ProteinMPNN, RoseTTAFold, AlphaFold and ESMFold.
- **Machine learning and artificial intelligence:** feature engineering, classification, model comparison, computer vision, deep learning, Transformers, generative models and protein language models. scikit-learn, Random Forest, SVM, gradient boosting, PyTorch and Hugging Face Transformers.
- **Programming and data analysis:** Python, R, Bash/Shell, SQL, pandas, NumPy, Biopython, exploratory data analysis, applied statistics and visualization with matplotlib and Plotly; library and command-line tool development.
- **Software development:** front end with React, JavaScript, HTML and CSS; back end with FastAPI and Flask; full stack development, REST APIs, GraphQL integration and scientific applications accessible through web interfaces.
- **Data engineering:** PostgreSQL, MySQL, SQLite, Redis, API integration, ETL pipelines, data modeling, batch ingestion and biological data organization with provenance tracking.
- **Workflows and high-performance computing:** Nextflow, Snakemake, Linux/Unix, Slurm and HPC; automation of omics analyses, execution of AI models and molecular simulations, resource management and reproducible workflows.
- **Quality and reproducibility:** Git/GitHub, Docker, Docker Compose, testing with pytest, technical documentation, version control and modular code organization.
- **Technical leadership and product management:** technical team management, Product Owner responsibilities, product lifecycle, requirements, MVPs, roadmaps, backlogs and stakeholders; Scrum, Kanban, Azure DevOps, sprint planning and delivery tracking.

PROFESSIONAL EXPERIENCE

Biotech Business

October 2025 to present

Bioinformatics Engineer and Head of Data Science

August 2026 to present

- **Technical leadership and delivery coordination:** translates scientific and business needs into requirements, priorities and software deliverables, connecting technical teams and research groups.
- **Computational solutions:** integrates bioinformatics, data engineering and machine learning into applications for biotechnology and agribusiness.
- **Product development:** contributes to roadmap definition, backlog prioritization and the ongoing development of solutions.
- **System and data integration:** develops APIs and database-connected applications to provide access to and exploration of biological information.

Bioinformatician

October 2025 to July 2026

- Developed computational solutions for organizing, processing and analyzing biological data for agribusiness.
- Developed machine learning and computer vision applications to support research analyses and processes.
- Implemented data integration routines, APIs and database queries.
- Collaborated with researchers and product teams to translate scientific needs into features and technical deliverables.

Seqera | Nextflow Ambassadors Program

Nextflow Ambassador

January 2026 to present | Volunteer role

- **Technical software collaboration:** works with developers in Seqera GitHub repositories, supporting the evolution of tools used in scientific workflows.
- **Testing and issue identification:** explores features, reproduces failures and identifies bugs during tool use.
- **Issue reporting and tracking:** documents problems, evidence and improvement suggestions to support assessment by development teams.
- **Reproducibility and workflows:** promotes good practices for pipeline development with Nextflow and methodologies from the nf-core community.

PickCells

March 2020 to August 2023 | Recife, Brazil

Innovation and Embedded Computer Vision (AI) Systems Development Specialist

August 2020 to August 2023

- **Computer vision for healthcare:** developed solutions for recognizing cellular patterns and identifying potentially malignant cells in hematological and cytological images, including cervical examinations and pleural fluid samples.
- **Software, AI and hardware integration:** developed embedded systems and automated devices connecting image acquisition, computational processing and machine learning model execution.
- **Prototyping and technical validation:** contributed to the construction, integration, testing and evaluation of IoT applications and devices for clinical image analysis.
- **Product Owner responsibilities:** defined and prioritized requirements and features for selected products, aligning user needs, business objectives and technical feasibility with development teams.
- **Product lifecycle management:** managed activities from needs identification and scope definition through MVP development, user validation and solution improvement based on results and feedback.
- **Agile management with Scrum and Azure DevOps:** organized and prioritized backlogs, planned sprints and tracked delivery, maintaining traceability between requirements, development activities and product evolution.
- **Roadmaps and prioritization:** organized product development around user value, technical dependencies and team delivery capacity.
- **Stakeholder engagement:** connected users, business teams and technical teams to clarify needs, negotiate priorities, present deliverables and incorporate feedback into product decisions.

Research, Technological Development and Innovation Intern

March 2020 to August 2020

- Contributed to the development of solutions for molecular diagnosis of SARS-CoV-2.
- Supported research, technological development and innovation activities for healthcare products.

Hospital das Clínicas da UFPE / EBSERH

Clinical Pathology Intern

October 2021 to March 2022 | Recife, Brazil

- Worked in sample screening, hematology, biochemistry, urinalysis, microbiology, hormone testing and serology.
- Participated in laboratory analyses and quality control procedures.
- Gained practical experience in sample processing and clinical laboratory testing workflows.

RESEARCH EXPERIENCE

Federal University of Minas Gerais | Institute of Biological Sciences

Bioinformatics Researcher | PhD

August 2024 to present | Belo Horizonte, Brazil

- **Computational method development:** builds tools for mining, identifying, characterizing and designing antimicrobial peptides.
- **Deep learning and generative models:** applies protein language models and sequence generation strategies to explore candidates with functional properties of interest.
- **Biological data integration:** analyzes sequences, physicochemical descriptors and structural information for candidate assessment and prioritization.
- **Molecular simulation:** prepares systems and performs molecular dynamics simulations to investigate stability, flexibility and peptide-membrane interactions.
- **Scientific software and reproducibility:** develops pipelines, applications and tools to make analyses reusable and accessible to other researchers.
- Conducts research on antimicrobial candidates and computational investigation of mechanisms associated with their activity.

Federal University of Pernambuco | Department of Genetics

Bioinformatics Researcher | Master’s degree

June 2022 to July 2024 | Recife, Brazil

- **AMPidentifier development:** built a machine learning tool for computational identification of antimicrobial peptides from sequences.
- **Model development and assessment:** prepared datasets, extracted physicochemical descriptors and compared algorithms for peptide classification.
- **Multi-omics analysis:** identified and computationally characterized five *Ricinus communis* defensins, integrating genomic and transcriptomic analyses with molecular modeling.
- **Functional characterization:** investigated gene expression and structural features related to defensin function.
- Developed computational routines to organize data, automate analyses and present reproducible research results.

Federal University of Pernambuco | Plant Genetics and Biotechnology Laboratory

Undergraduate Research Fellow

January 2020 to May 2022 | Recife, Brazil

- Conducted bioinformatics research on the identification and characterization of bioinspired antimicrobial peptides.
- Investigated eIF4E family sequences in *Vigna* species and their relationship with defense mechanisms against pathogens.
- Integrated sequence analysis with functional and structural information to investigate molecules of biotechnological interest.

Oswaldo Cruz Foundation | Aggeu Magalhães Institute

Undergraduate Research Fellow

November 2016 to December 2019 | Recife, Brazil

- **Structural biology and computational chemistry:** worked on molecular modeling and investigation of protein structural properties and interactions.
- **Protein engineering:** contributed to research on molecular characterization and development for potential diagnostic and vaccine applications.
- **Virology research:** conducted activities in the Department of Virology and Experimental Therapy, connecting biological questions with computational approaches.
- Analyzed and interpreted molecular models to support research hypotheses and decisions.

Federal University of Pernambuco | Keizo Asami Institute

Undergraduate Research Fellow

May 2015 to December 2016 | Recife, Brazil

- Participated in molecular biology, human genetics and bioinformatics research.
- Investigated ancestry markers and phenotype prediction in forensic genetics projects.
- Collaborated on activities with the Forensic Genetics Institute of the Scientific Police of Pernambuco.

SOFTWARE AND BIOINFORMATICS PROJECTS

Applications and tools

AMPidentifier | Antimicrobial peptide classification

- Classifies FASTA sequences, returning predicted labels and probabilities with a decision threshold of 0.90.
- Integrates Biopython, modlAMP and scikit-learn, with Random Forest, ensemble voting and support for external models.
- Available as a command-line tool, PyPI package and Flask web application served by Gunicorn on Render.
- **Result:** AUC-ROC of 0.950 on an external benchmark. **INPI software registration:** BR 51 2025 005859-4.

decryptAMP | Cryptic peptide mining

- Sliding-window mining of RefSeq proteomes, with charge and hydrophobicity filtering and hash-based deduplication.
- Implemented in Python with Biopython, NumPy and pandas, with parallel processing by proteome.
- **Processing scale:** 105 proteomes from six kingdoms and 1.28 million nonredundant candidates. Output integrated with AMPidentifier for candidate classification.

ecAMPdb | Cryptic antimicrobial peptide integration and exploration

- Data warehouse integrating sequences, physicochemical descriptors and provenance: source proteome, taxon, precursor coordinates and model-assigned probability.
- Flask back end with PostgreSQL, versioned batch ingestion, sequence and descriptor-range searches, and an embedding atlas.
- **Core functionality:** combined queries by taxonomic origin and physicochemical properties.

AMPcraft | Computational peptide design

- Peptide design strategies built on ESM-2 with PyTorch and Hugging Face Transformers.
- Fine-tuning with layer freezing and prediction heads for activity, hemolysis and cytotoxicity.
- Generation through masked-position sampling, attention-guided refinement and three-dimensional modeling with ESMFold.
- **Approach:** uses hemolysis and cytotoxicity predictions as constraints during candidate generation.

BILBO | Lipid bilayer construction

- Assembles bilayers from Martini 3 topologies and exports files for GROMACS simulation preparation.
- Developed in Python with MDAnalysis and provided as a Flask web application.
- Builds systems from a declared lipid composition, enabling taxon-specific membrane models.
- Documentation and file organization support reproducible simulation systems.

Pure Design | Design system for interfaces and scientific figures

- Shared design system for application interfaces and scientific visualizations.
- Tokens organized in tokens.css and patterns.css, with scripted checks for contrast against WCAG criteria and spacing-scale consistency.

- Corresponding themes for matplotlib and Plotly. **Goal:** visual consistency across applications, charts and publication figures.

GenVar Dashboard | Gene and variant exploration

- Web application for integrated queries across genes, variants, functional annotations, population frequencies and protein structures.
- Integrates Ensembl, gnomAD, ClinVar, AlphaFold and UniProt, alongside scores provided through MyVariant.info.
- Architecture includes a React front end, FastAPI API, Redis cache, Docker containers and pytest tests.
- Developed as part of the MBA in Software Engineering at the University of São Paulo (USP).

GETVar | Variant processing and annotation

- Application for processing VCF files and annotating genetic variants.
- Integrates workflows with sources including dbSNP, ClinVar and Ensembl. Implemented with Python, Flask, Snakemake and Bootstrap.

Plant eIF4E Atlas | Biological data engineering

- Platform for integrating and exploring information on plant eIF4E proteins.
- ETL pipeline and data model for taxonomic and functional queries, using Python, SQL, APIs and web visualization.
- Developed as part of the postgraduate specialization in Data Science and Analytics at the Pontifical Catholic University of Rio de Janeiro (PUC-Rio).

Python packages

ampidentifier | Available on PyPI

- AMPidentifier engine API through run_prediction_pipeline, supporting the internal model, ensemble and external models.
- Developed for Python 3.8 or later with scikit-learn, Biopython, NumPy, pandas and modlAMP. Usable in scripts, Jupyter and Google Colab.
- Automated publishing through GitHub Actions triggered by version tags.

decryptamp | In preparation

- Programmatic interface for cryptic peptide mining in FASTA files, with descriptor filtering, deduplication and export for integration with ampidentifier.
- Implemented with Biopython, NumPy and pandas. Based on completed processing of 105 proteomes and 1.28 million nonredundant sequences.

EDUCATION

PhD in Bioinformatics

Federal University of Minas Gerais - UFMG

2024 to present.

MBA in Software Engineering

University of São Paulo - USP

In progress. Expected completion: **December 2026**.

Postgraduate Specialization in Data Science and Analytics

Pontifical Catholic University of Rio de Janeiro - PUC-Rio

Completed in 2026.

Master's degree in Genetics and Molecular Biology

Federal University of Pernambuco - UFPE

Completed in 2024. Research focused on bioinformatics.

Bachelor's degree in Biomedical Sciences

Federal University of Pernambuco - UFPE

Completed in 2022.

Technical Diploma in Software Development

Escola Técnica Estadual de Pernambuco - ETEPE

Completed in 2013.

SELECTED CERTIFICATIONS AND ADDITIONAL TRAINING

- **Nextflow Training** | Seqera | September 2025.
- **Agile Project Management Professional Certificate** | Atlassian | May 2025.
- **Generative AI** | MIT Professional Education | April 2025.
- **Requirements Engineering and Agile Product Management** | PUC-Rio | April 2025.
- **Microsoft Azure AI Essentials: Workloads and Machine Learning** | March 2025.
- **Advanced Gemini for Developers** | June 2024.
- **Career Essentials in GitHub Professional Certificate** | January 2024.
- **Multidimensional Arrays and Their Traversal in Python** | CodeSignal | August 2022.
- **Slurm Workload Manager** | National Laboratory for Scientific Computing, Brazil | June 2022.

COMMUNITY AND TEACHING

Seqera | Nextflow Ambassadors Program

Student Ambassador

January 2026 to present | Volunteer

- Participates in the scientific workflow community, promoting reproducibility, collaboration and Nextflow adoption.
- Shares knowledge on bioinformatics pipeline development and organization.
- Engages with researchers and developers to support the use of tools and practices from the Nextflow and nf-core communities.

Invited teaching and educational materials

- **PyMOL teaching materials for the Basic Bioinformatics Course** | 2026.
- **Molecular Docking & Dynamics: Breathing Motion into Life’s Building Blocks** | First and second editions | 2025.
- **Molecular Modeling with Machine Learning Techniques** | First and second editions | 2025.
- **Bioinformatics: A Theoretical-Practical Approach** | 2024.
- **Introduction to Bioinformatics: From DNA to Proteins** | Catholic University of Pernambuco | 2022.

Teaching assistantships | Federal University of Pernambuco

- **Human Genetics** | 2017 to 2018.
- **Molecular Tools Applied to Clinical Diagnosis** | 2017 to 2018.

PUBLICATIONS

Journal articles

The Regulatory Army of Plant Defense: Transcription Factors in the War for Plant Immunity

José Ribamar Costa Ferreira-Neto; Agnes Angélica Guedes de Barros; Ana Luíza Trajano Mangueira de Melo; Lidiane Lindinalva Barbosa Amorim; **Madson Allan de Luna Aragão**; João Pacífico Bezerra-Neto; Laiane Silva Maciel; Manassés Daniel da Silva; Paulo Vitor Galdino da Silva; Ana Maria Benko-Iseppon.

International Journal of Molecular Sciences, 27(16), 7315, 2026. DOI: 10.3390/ijms27167315

Unveiling Three Functionally Diverse Isoforms of eIF4E in Cowpea Through a Multi-Omics Approach

Madson Allan de Luna-Aragão; Fernanda Alves de Andrade; Saulo Rafael Mendes Penna; Laiane Silva Maciel; Laura Maria Rodrigues-Paixão; Ayug Bezerra Lemos; José Diogo Cavalcanti Ferreira; Francisco José Lima Aragão; Valesca Pandolfi; Ana Maria Benko-Iseppon.

Agronomy, 16(7), 766, 2026. DOI: 10.3390/agronomy16070766

Transposable elements: Functional aspects and applications as drivers of crop innovation

Flávia Layse Belém Medeiros; José Ribamar Costa Ferreira-Neto; Jarbson Henrique Oliveira Silva; Cinthia Carla Claudino Grangeiro Nunes; Ana Luíza Trajano Mangueira de Melo; Ayug Bezerra Lemos; **Madson Allan de Luna-Aragão**; Agnes Angélica Guedes de Barros; Simon Orozco-Arias; Eliseu Binneck; Francismar Correa Marcelino-Guimarães; Ana Cristina Brasileiro-Vidal; Ana Maria Benko-Iseppon.

Crop Science, 66(2), e70257, 2026. DOI: 10.1002/csc2.70257

Deciphering Cowpea Resistance to Potyvirus: Assessment of eIF4E Gene Mutations and Their Impact on the eIF4E-VPg Protein Interaction

Fernanda Alves de Andrade; **Madson Allan de Luna-Aragão**; José Diogo Cavalcanti Ferreira; Fernanda Freitas Souza; Ana Carolina da Rocha Oliveira; Antônio Félix da Costa; Francisco José Lima Aragão; Carlos André dos Santos-Silva; Ana Maria Benko-Iseppon; Valesca Pandolfi.

Viruses, 17(8), 1050, 2025. DOI: 10.3390/v17081050

Omics-driven bioinformatics for plant lectins discovery and functional annotation - A comprehensive review

Ruana Carolina Cabral da Silva; Ricardo Salas Roldan-Filho; **Madson Allan de Luna-Aragão**; Roberta Lane de Oliveira Silva; José Ribamar Costa Ferreira-Neto; Manassés Daniel da Silva; Ana Maria Benko-Iseppon.

International Journal of Biological Macromolecules, 279, Part 4, 135511, 2024. DOI: 10.1016/j.ijbiomac.2024.135511

Association strength of E6 to E6AP/p53 complex correlates with HPV-mediated oncogenesis risk

Matheus Vitor Ferreira Ferraz; Isabelle Freire Tabosa Viana; Danilo Fernandes Coêlho; Carlos Henrique Bezerra da Cruz; Maíra de Arruda Lima; **Madson Allan de Luna Aragão**; Roberto Dias Lins.

Biopolymers, 113(10), e23524, 2022. DOI: 10.1002/bip.23524

Book chapter

Approaches for Identification and Validation of Antimicrobial Compounds of Plant Origin: A Long Way from the Field to the Market

Lívia Maria Batista Vilela; Carlos André dos Santos-Silva; Ricardo Salas Roldan-Filho; Pollyanna Michelle da Silva; Marx de Oliveira Lima; José Rafael da Silva Araújo; Wilson Dias de Oliveira; Suyane de Deus e Melo; **Madson Allan de Luna Aragão**; Thiago Henrique Napoleão; Patrícia Maria Guedes Paiva; Ana Christina Brasileiro-Vidal; Ana Maria Benko-Iseppon.

Eco-Friendly Biobased Products Used in Microbial Diseases. CRC Press, p. 183-222, 2022. DOI: 10.1201/9781003243700-12

Publication profiles: ORCID | Google Scholar.

AWARDS AND HONORS

- **Next Generation Bioinformatician (NGB) |** 21st International Conference of the Brazilian Association for Bioinformatics and Computational Biology **|** X-Meeting 2025.
- **Travel Grant Award |** School on Biological Physics and Biomolecular Simulations in the Machine Learning Era **|** ICTP-SAIFR.
- **Honorable Mention in Human and Forensic Genetics |** CNPq / Federal University of Pernambuco.
- **Young Geneticist Award of the Northeast |** XXI Northeast Genetics Meeting **|** ENGNE.
- **Honorable Mention |** Postgraduate Genetics Meeting **|** Federal University of Pernambuco.
- **Best Poster Award |** XIII Genetics and Molecular Biology Program Meeting.
- **Travel Grant Award |** AI for Protein Design (AI4PD) **|** The Protein Society.
- **Certificate of Excellence in Peer Reviewing |** Elsevier.
- **Highest Admission Score, Master’s Program in Genetics and Molecular Biology |** Federal University of Pernambuco **|** Ranked first overall among all accepted applicants.
- **Highest Admission Score, PhD Program in Bioinformatics |** Federal University of Minas Gerais **|** Ranked first overall among all accepted applicants.
- **Honorable Mention, Pernambuco Chemistry Olympiad (OPEQ) |** National Chemistry Olympiad Program and Pernambuco Science Space.

LANGUAGES

Portuguese: native.

English: advanced.